

Technical Demo: Graph Representation Learning for Geography

Yuta Sato

University of Liverpool

May 7th, 2025

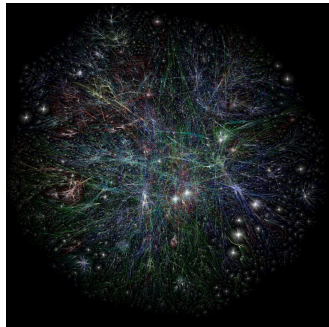
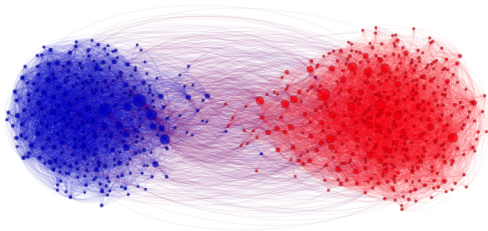
Agenda

- What is graph?
- Examples of Graphs in Geography
- What is Graph Representation Learning
- Why GNNs?
- Message Passing Framework
- Notable GNN Models
- GNNs in Geographic Data Science
- Technical Demonstration

Note: This presentation provides an overview and does not delve deeply into Graph Theory, Network Science, or specific Deep Learning implementation details.

What is a Graph? (1/2)

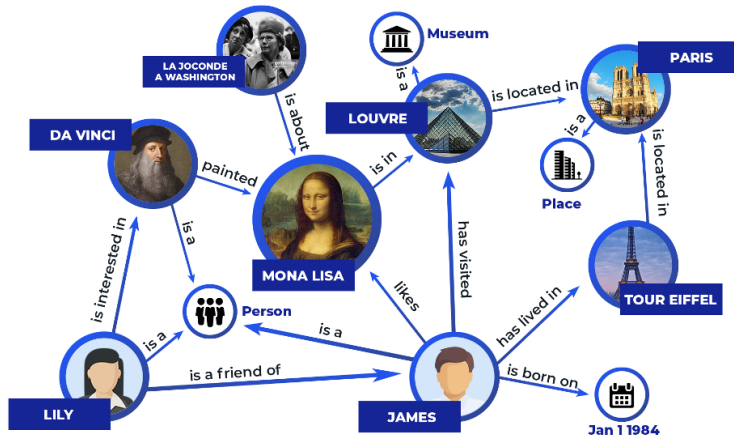
- A graph is a mathematical structure consisting of:
 - **Nodes (vertices)**: represent entities
 - **Edges**: represent relationships between entities
- Real-world examples:
 - **Social networks**: Users (nodes) and friendships (edges)
 - **Molecules**: Atoms (nodes) and bonds (edges)
 - **Internet**: Global IP address (nodes) and web links (edges)



What is a Graph? (2/2)

- **Heterogeneous Graphs:**

- Multiple types of nodes and/or edges
- Different node/edge types may have different feature structures



Examples of Graphs in Geography (1/3): Streets

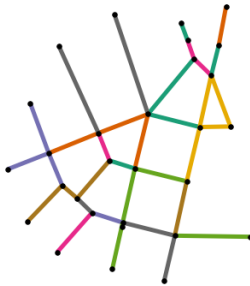
• Street Networks:

- In general planar and can be represented as primary and dual graphs (Porta et al., 2006)
- Primal Graph (nodes: intersections/segments, edges: streets)
- Dual Graph (nodes: road segments, edges: intersections)

Road network



Primal graph



Dual graph



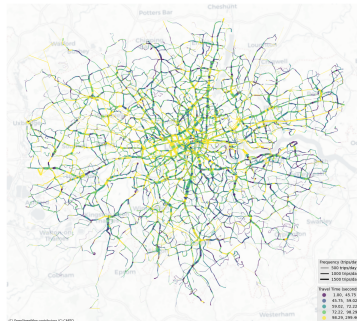
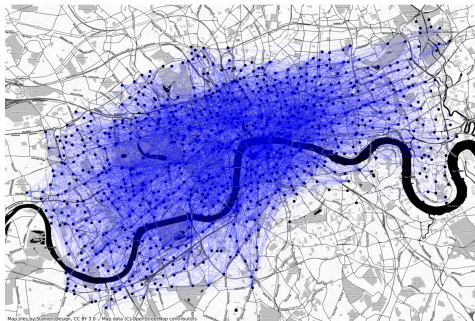
Examples of Graphs in Geography (2/3): Mobility

- **Origin-Destination (OD) Matrices:**

- Nodes: locations/zones, edges: movement flows
- Weights: trip frequency/volume, mode share

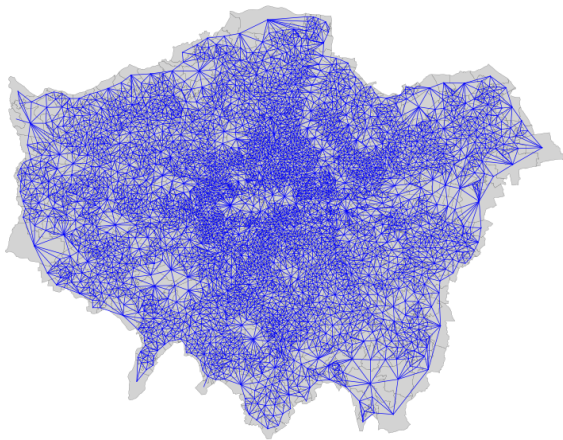
- **Transportation Networks:**

- Nodes: locations/zones, edges: transportation routes
- Weights: distance, travel time, cost



Examples of Graphs in Geography (3/3): Spatial Contiguity / Proximity

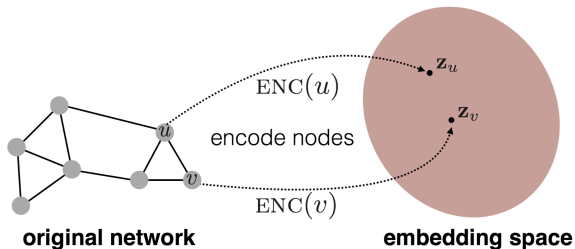
- Spatial Contiguity/Proximity Networks:
 - Regions/areas (nodes) connected by shared boundaries or proximity (edges)



What is Graph Representation Learning? (1/2)

- **Graph Representation Learning:**

- Machine learning models that maps nodes/graphs to vectors in d -dimensional embedding space (Euclidean)
- Node prediction ($f_n : \{\mathcal{G}_i\} \rightarrow \mathbb{R}^{n \times d}$)
- Edge prediction ($f_e : \{\mathcal{G}_i\} \rightarrow \mathbb{R}^{n \times n \times d}$)
- Graph prediction ($f_g : \{\mathcal{G}_i\} \rightarrow \mathbb{R}^d$)

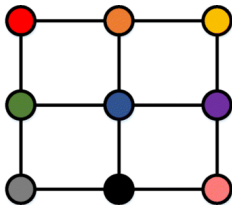


What is Graph Representation Learning? (2/2)

- **Graph Neural Networks (GNNs):**
 - Deep learning models for graph-structured data
 - Learn representations through **message passing framework**
 - Update node features by aggregating neighbor information

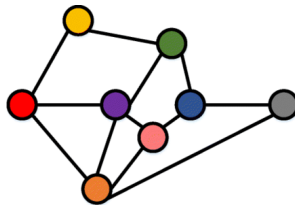
Why GNNs?

- Traditional ML struggles with graph data:
 - Fixed-size input assumption (e.g., images, sequences)
 - Order-dependent (graphs are inherently unordered)
 - Fails to capture relational structure explicitly
- GNNs offer key advantages:
 - Handle irregular, non-Euclidean graph structures
 - Permutation invariant/equivariant (node order doesn't matter)
 - Leverage node features, edge features, and topology simultaneously



CNN

In Euclidean Space



GNN

In Non-Euclidean Space

Message Passing Framework

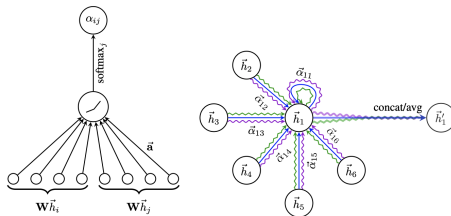
- **Core idea:** Nodes iteratively aggregate information from neighbors to update their own representations.
- **General Formulation** (Layer k to $k + 1$):

$$\mathbf{h}_v^{(k+1)} = \gamma^{(k)} \left(\mathbf{h}_v^{(k)}, \square_{u \in \mathcal{N}(v)}^{(k)} \phi^{(k)} \left(\mathbf{h}_v^{(k)}, \mathbf{h}_u^{(k)} \right) \right)$$

- $\mathbf{h}_v^{(k)}$: Representation (hidden state) of node v at layer k .
- $\mathcal{N}(v)$: Set of neighbors of node v .
- $\phi^{(k)}$: Message function (computes message from neighbor u to v). Often involves learnable weights $\mathbf{W}^{(k)}$.
- $\square^{(k)}$: Aggregation function (combines messages from neighbors, e.g., sum, mean, max). Must be permutation invariant.
- $\gamma^{(k)}$: Update function (combines aggregated message with node's previous state, often involves activation). E.g., MLP + Activation($\sigma(x) = \frac{1}{1+e^{-x}}$, ReLU= $\max(0, x)$).

Notable GNN Models (1/3)

- **Graph Convolutional Networks (GCNs)** (Kipf & Welling, 2016a):
 - Aggregates neighbor features equally (often mean aggregation).
- **Graph Attention Networks (GATs)** (Veličković et al., 2017):
 - Assigns different importance (attention weights) to neighbors during aggregation.



Notable GNN Models (2/3)

- **GraphSAGE** (Hamilton et al., 2017):

- Scalable GNN that samples a fixed-size neighborhood for each node.

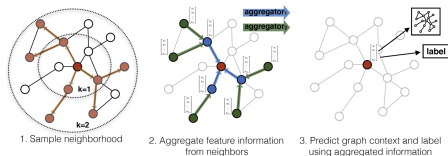


Figure 1: Visual illustration of the GraphSAGE sample and aggregate approach.

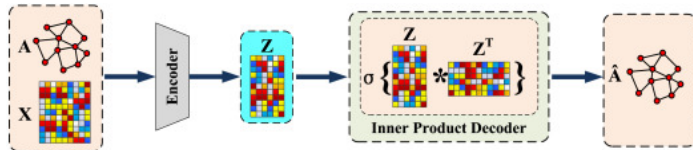
- **Graph Transformers** (Yun et al., 2019):

- Uses self-attention over all nodes (or sampled nodes) to capture global dependencies, inspired by NLP Transformers.

Notable GNN Models (3/3)

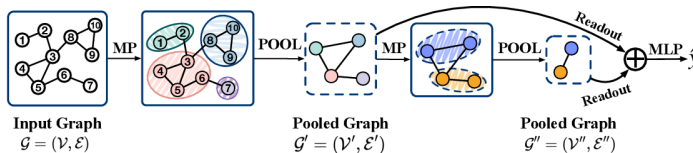
- **Graph Autoencoder (GAE)** (Kipf & Welling, 2016b):

- Unsupervised: Learn compressed node embeddings by encoding the graph structure (and features) and decoding to reconstruct it (often the adjacency matrix).



- **Graph Pooling** (Liu et al., 2023):

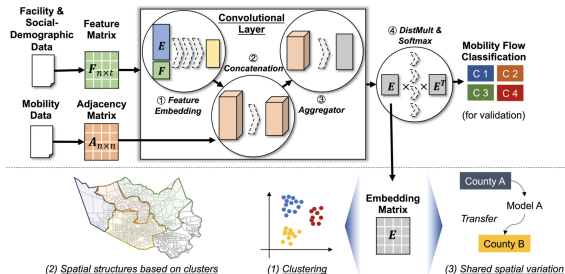
- Down-samples the graph (reduces nodes) to create coarser representations, analogous to pooling in CNNs. Hierarchical or global.



GNNs in Geographic Data Science (1/3)

- **Example: Urban Structure Embedding** (Fan et al., 2024)

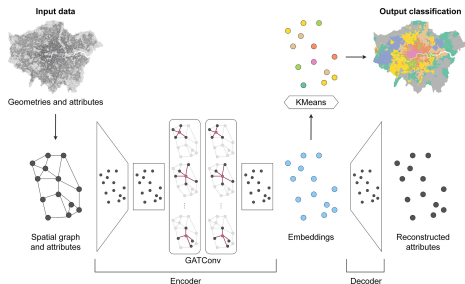
- **Data:** OD matrix (mobility flows) for 16 US cities (nodes: 4km $\times$ 4km grid cells, edges: OD).
- **Features:** Node: Census demographics, POIs. Edge: Trip counts.
- **Task:** Link prediction (mobility flow) using GCN. Node embeddings were clustered by Hierarchical Clustering to reveal functional urban zones.
- **Finding:** GNN embeddings captured spatial structures related to urban functions (residential, commercial etc.).



GNNs in Geographic Data Science (2/3)

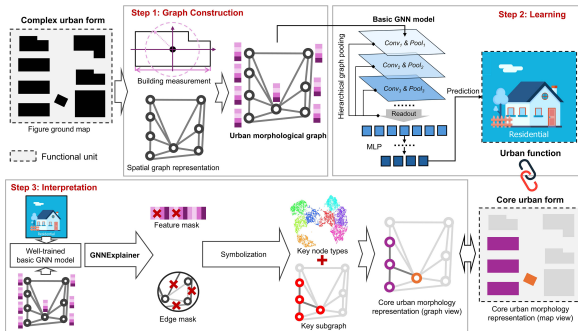
- **Example: Geodemographic Clustering** (Sabbata & Liu, 2023)

- **Data:** Greater London's administrative boundary in 2011 UK Census (nodes: Output Areas, edges: Queen-contiguity / KNN / distance threshold)
- **Features:** Node: Census variables (167 z-scores, 60 k-vars). Edge: None
- **Task:** Geodemographic clustering by GAE with GCN / GAT layers (Node embeddings clustered by k-means and spatial fuzzy c-means)
- **Finding:** GAE performed similarly to classic approaches on class homogeneity metrics while providing higher spatial clustering



GNNs in Geographic Data Science (3/3)

- **Example: Predicting Urban Functions by Forms** (Chen et al., 2025)
 - **Data:** Boston (nodes: 98,331 buildings, edges: Delaunay triangulation)
 - **Features:** Node: morphological metrics (size, shape, orientation, density), edge: distances
 - **Task:** Predict urban functions using urban morphological graphs
 - **Finding:** Proposed model effectively revealed high-order relationships between urban forms and functions



- Implementing GNNs with PyTorch Geometric (PyG):
 - Popular library built on PyTorch.
 - Efficient handling of sparse graph data.
 - Provides standard GNN layers (GCN, GAT, etc.) and benchmark datasets.



- Demo Outline:
 - Loading a sample graph dataset (e.g., Cora citation network or a simple geographic one).
 - Defining a GCN model using PyG layers.
 - Training the model for a node classification task.
 - Visualizing node embeddings (optional).
- Demo Link: [Click here](#)

References I

- Chen, D., Feng, Y., Li, X., Qu, M., Luo, P., & Meng, L. (2025). Interpreting core forms of urban morphology linked to urban functions with explainable graph neural network. *Computers, Environment and Urban Systems*, 118, 102267.
<https://doi.org/https://doi.org/10.1016/j.compenvurbsys.2025.102267>
- Fan, C., Yang, Y., & Mostafavi, A. (2024). Neural embeddings of urban big data reveal spatial structures in cities. *Humanities and Social Sciences Communications*, 11, 409.
<https://doi.org/10.1057/s41599-024-02917-6>
- Hamilton, W. L., Ying, R., & Leskovec, J. (2017). Inductive representation learning on large graphs. *CoRR*, abs/1706.02216. <http://arxiv.org/abs/1706.02216>
- Kipf, T. N., & Welling, M. (2016a). Semi-supervised classification with graph convolutional networks. *arXiv preprint arXiv:1609.02907*.
- Kipf, T. N., & Welling, M. (2016b). Variational graph auto-encoders.
<https://arxiv.org/abs/1611.07308>

References II

- Liu, C., Zhan, Y., Wu, J., Li, C., Du, B., Hu, W., Liu, T., & Tao, D. (2023). Graph pooling for graph neural networks: Progress, challenges, and opportunities.
<https://arxiv.org/abs/2204.07321>
- Porta, S., Crucitti, P., & Latora, V. (2006). The Network Analysis of Urban Streets: A Primal Approach. *Environment and Planning B: Planning and Design*, 33(5), 705–725.
<https://doi.org/10.1068/b32045>
- Sabbata, S. D., & Liu, P. (2023). A graph neural network framework for spatial geodemographic classification. *International Journal of Geographical Information Science*, 37(12), 2464–2486. <https://doi.org/10.1080/13658816.2023.2254382>
- Veličković, P., Cucurull, G., Casanova, A., Romero, A., Lio, P., & Bengio, Y. (2017). Graph attention networks. *arXiv preprint arXiv:1710.10903*.
- Yun, S., Jeong, M., Kim, R., Kang, J., & Kim, H. J. (2019). Graph transformer networks. *Advances in neural information processing systems*, 32.